BrewBot: Human-State-Aware Multimodal Interaction for Robotic Drink Assistance

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Figure 1: BrewBot offering and serving a drink in the lab kitchen. The arm hangs from a ceiling rail, so it can travel between the sink, the coffee machine and the person it is serving.

Abstract

Physiological sensing enables robots to adapt their behavior to aspects of a user's current state without requiring an explicit request. In proactive human–robot interaction, however, state inference is only part of the challenge: a robot may initiate an interaction before the user has explicitly invited it, making interruption, user control, and the choice of interaction modality central design concerns. We present BrewBot, a robotic drink assistant built around a ceiling-mounted Kinova Gen3 arm in a shared lab kitchen. BrewBot combines heart rate, electrodermal activity, skin temperature, and motion cues in a rule-based estimator to derive coarse-grained candidate user states and corresponding beverage suggestions. Importantly, physiological inference determines a suggestion rather than an autonomous action: the user remains in control of whether and how the interaction proceeds. Suggestions can be accepted, rejected, or changed through speech or gestures. An open-palm gesture can interrupt verbal interaction and transfer the remaining dialogue to an embodied gesture mode in which the robot mimes beverage options and receives thumbs-up or thumbs-down responses. BrewBot subsequently prepares or fetches the selected drink using the robot arm and smart-home infrastructure, with collaborative interaction required for non-automated tasks such as filling a glass at the tap. BrewBot as an integrated design case rather than an empirically evaluated HRI system and discuss three design considerations emerging from its implementation: separating state inference from interruption policy, treating modalities as substitutable rather than merely parallel, and preserving user agency under proactive assistance.

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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Robotic planning*.

Keywords

human-robot interaction, proactive robots, physiological sensing, multimodal interaction, gesture interaction, service robotics, ROS 2

1 Introduction and Related Work

A service robot is typically reactive: the human makes a request and the robot responds. Physiological and contextual sensing enables a different form of interaction. Rather than waiting until a person explicitly asks for assistance, a robot can use cues about the person's current physical or affective state to decide that assistance might be appropriate and proactively offer it. This idea is particularly relevant to human–robot interaction (HRI), where assistance may involve physical action in a shared environment rather than only information or recommendations.

Physiological signals such as heart rate and electrodermal activity (EDA) are commonly used as indicators for autonomic arousal and stress [3, 4]. Wearable devices make such signals continuously available outside tightly controlled laboratory setups, although substantial inter-individual variation and limitations in signal quality remain [11, 12]. This relates to Schmidt's concept of implicit interaction, in which systems respond to contextual cues that the user did not deliberately produce as commands [10]. Human-state-aware interaction therefore offers an opportunity for robots to adapt without requiring the user to repeatedly articulate their needs.

Proactivity introduces an additional problem, however. A command-driven robot is normally addressed at a time selected by the human. A proactive robot instead chooses when to initiate an interaction. Research on proactive HRI consequently emphasizes not only whether robots should take initiative, but also the circumstances in which such initiative is desirable [1, 5]. A potentially useful suggestion can still be disruptive if the user is occupied, already interacting with another person, unable to speak, or simply uninterested. For a state-aware robot, estimating *what* might be useful and deciding *when and how* to offer it are therefore separate questions.

This becomes especially relevant when interaction is multimodal. Speech is convenient when the user is free to converse, but it is not always the cheapest or most appropriate response channel. Multimodal interaction research has long argued against treating modalities as interchangeable copies of one another and shows that users select modalities according to task and situational constraints [9]. In human-agent collaboration, related mixed-initiative perspectives emphasize that initiative and control may shift between human and system rather than belonging exclusively to one side [2]. For a proactive robot, multimodality can thus serve not only recognition robustness but also *user control over the interaction itself.*

We investigate these questions through BrewBot, an integrated robotic drink assistant. BrewBot combines physiological sensing, proactive beverage suggestions, speech, human hand gestures, robot gestures, smart-home devices, and physical manipulation in a shared kitchen environment. The beverage-serving task provides a concrete setting in which inferred human state can influence robot behavior while every suggestion still requires explicit human acceptance. We chose drink assistance because it turns an inferred state into a concrete, low-stakes offer: an unsuitable suggestion can be declined, changed, or ignored without serious consequence.

The project makes three main contributions as a design case. First, BrewBot integrates wearable and heart-rate sensing into an end-to-end robotic assistance pipeline in which approximate physiological state estimates inform context-dependent beverage suggestions. Second, users can accept, reject, or change suggestions through speech or gestures and can explicitly replace an ongoing verbal exchange with an embodied gesture-based interaction. Third, BrewBot connects state inference and interaction management to real physical actions, including drink preparation through smart-home infrastructure, delivery to different locations, and collaborative execution of tasks that cannot be fully automated.

BrewBot was developed by a three-person student team around an intensive on-site practical phase, with preparatory work on sensing, arm control, and the kitchen setup before the final integration. The prototype was refined through repeated demonstration runs rather than through a formal study. These runs were used to test whether the complete interaction could proceed from sensed state to suggestion, human confirmation, and physical drink delivery, and they exposed the interruption, modality-switching, and legibility issues discussed below.

Our goal is not to claim validated inference of psychological states or superior user experience. No controlled user study was conducted. Instead, BrewBot is presented as an integrated prototype from which we derive design considerations and testable hypotheses for future state-aware HRI systems.

2 BrewBot System and Interaction Design

2.1 Architecture and Human-State Inference

BrewBot operates in an instrumented lab kitchen at LMU Munich. The robotic platform consists of a ceiling-mounted Kinova Gen3 six-degree-of-freedom arm with a Robotiq 2F-140 gripper. The arm is attached to a motorized linear rail that allows it to travel between relevant areas of the kitchen, including the workstation, sink, and networked coffee machine. The system is implemented in ROS 2 [7].

Physiological sensing is distributed across a dedicated heart-rate sensor and an Empatica E4 wristband. The E4 provides electrodermal activity, skin temperature, and acceleration, while the separate heart-rate sensor directly reports heart rate in beats per minute. Cameras in the environment and on the robot support gesture recognition. Speech input is transcribed locally, speech output is produced on site, and a locally hosted language model supports the interpretation of free-form replies. Home Assistant connects BrewBot to smart-home infrastructure including the coffee machine and lights. In the implemented system, these functions are distributed across ROS 2 nodes for sensing, state estimation, triggering, interaction management, speech input and output, gesture recognition, arm control, and smart-home actuation. This structure matters for the interaction as well as the architecture: sensing, interruption, dialogue, and physical action can be adjusted independently. When non-essential components fail, the system can still preserve the user's control over whether a drink is prepared or delivered.

Conceptually, BrewBot follows a sense–infer–interact–act architecture (Figure 2). A central design decision is that the state estimator itself remains passive. It maintains the current sensor-derived state and corresponding beverage suggestion but does not command the robot to approach or interrupt the user. A separate trigger and interaction layer decides when a state estimate should result in a proactive offer. This distinction separates two questions with fundamentally different inputs:

What might be appropriate to offer? This depends primarily on physiological and activity-related cues.

Is it appropriate to initiate an interaction now? This additionally depends on interaction history, whether the user is present, whether another interaction is already active, and whether the same state has already triggered an offer recently.

The estimator uses a deliberately simple rule-based approach. Skin temperature, motion, heart rate relative to a configured baseline value, and EDA relative to a running baseline are combined into approximate user-state categories. Thermal and activity cues are considered before arousal-related cues so that obvious physiological confounds, such as an elevated heart rate during movement, are less likely to be interpreted directly as an affective state. The resulting internal states include *hot*, *cold*, *active*, *stressed/elevated arousal*, *fatigued/low arousal*, and *calm*. They map to illustrative beverage policies such as water for heat or activity, tea for cold or elevated arousal, and coffee for low-arousal conditions.

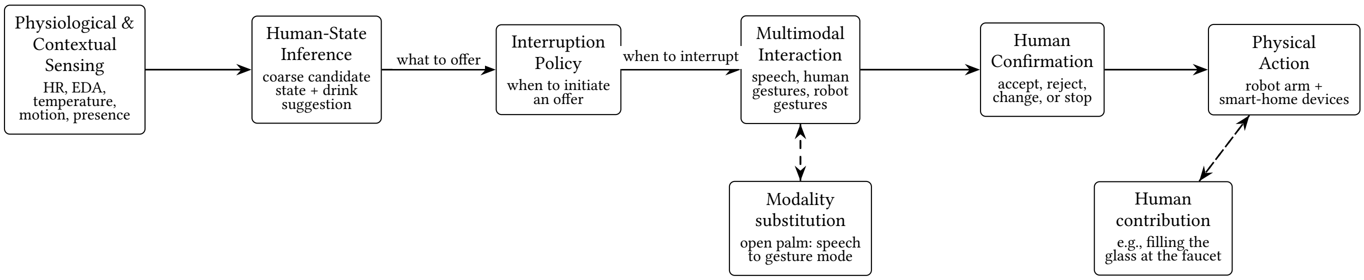


Figure 2: Conceptual architecture of BrewBot. Physiological and contextual signals are transformed into a candidate user state. The state estimator informs, but does not itself initiate, interaction. A separate interaction policy determines whether a suggestion should be made. Users can then respond through speech or gestures before the accepted action is transferred to the robot and smart-home components. Dashed feedback arrows illustrate modality substitution and the human contribution required during collaborative tasks such as filling a glass at the tap.

These labels must be understood as operational categories used by the prototype rather than validated diagnoses. Heart rate or EDA alone do not establish that a person is stressed, nor does reduced EDA establish fatigue. The mapping between physiological cues, inferred state, and drink should therefore be interpreted as a demonstrator for adaptive interaction rather than as a medical or physiological recommendation.

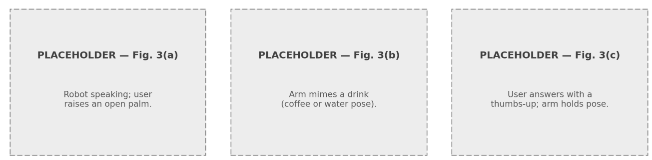
Heart-rate variability (HRV) was considered during development because of its established relation to stress [4]. Reliably deriving beat-to-beat intervals from the E4's wrist photoplethysmography, however, introduced a signal-processing problem beyond the scope of the project. BrewBot therefore uses a separate heart-rate sensor and retains the E4 primarily for EDA, skin temperature, and motion. This pragmatic division also reflects published validation work showing that physiological measures from wrist-worn devices are sensitive to measurement conditions and signal quality [11, 12].

To prevent momentary fluctuations from immediately producing unsolicited interaction, BrewBot requires persistent evidence of a relevant state change and applies a cooldown after proactive suggestions. This is not intended to solve the broader problem of interruption timing, but it avoids a simple failure mode in which noisy or persistent physiological conditions cause repeated offers.

2.2 Proactive and Multimodal Interaction

BrewBot treats physiological inference as a basis for offering assistance, not as an authorization to act. Even when BrewBot derives a candidate state and corresponding beverage, it does not automatically prepare the drink. Instead, the robot makes an offer and waits for confirmation. The user may accept, reject, or request another available beverage. This distinction preserves human agency despite the proactive nature of the system.

The default interaction channel is verbal. BrewBot can address the user, explain or contextualize its suggestion, and ask whether the proposed drink is desired. Speech recognition converts the reply to text, after which a locally hosted language model classifies the response into a restricted set of meaningful outcomes. This allows responses that are more expressive than simple yes/no answers. For example, a reply such as "No, but I'd rather have coffee" can be interpreted simultaneously as rejection of the suggestion and selection of an alternative beverage.



(a) **(b)** **(c)**

Figure 3: Modality substitution in BrewBot. (a) The user raises an open palm during a verbal offer, stopping the speech channel. (b) The robot continues the interaction by physically miming the available beverage options. (c) The user accepts or rejects the currently presented option with a thumbs-up or thumbs-down. The interaction therefore continues after the user has rejected the original communication modality.

The language model is intentionally kept outside the robot's physical control path. It can generate or interpret language but cannot introduce unsupported drinks or directly select robot motions. If language processing fails, BrewBot falls back to predefined prompts and responses. This confines the least deterministic component of the system to communication rather than physical control.

Human hand gestures provide a second reply channel. MediaPipe [6] is used to recognize three deliberately restricted gestures: thumbs up, thumbs down, and open palm. Recognition is stabilized across consecutive frames before a gesture is accepted. Gesture semantics depend on the interaction context rather than being defined globally. A thumbs-up may indicate acceptance of a drink offer, while during another task it can indicate that a human contribution has been completed.

The "open-palm" gesture illustrates this context-dependent design. During verbal interaction, an open palm immediately stops ongoing speech. It then publishes a stop signal for text-to-speech and marks the drink request so that the arm controller offers the menu through predefined mime motions. The gesture therefore rejects the current communication modality, but not necessarily the underlying offer. BrewBot can then transfer the remaining interaction from speech to an embodied gesture mode.

In gesture mode, the robot represents available beverages through predefined arm and gripper motions. It orients toward a predefined user-facing pose, performs the corresponding drink gesture, and waits for feedback. A thumbs-up selects the presented drink; a thumbs-down proceeds to another option; an additional open-palm response can end the offer.

This distinction leads to an important design observation: *rejecting a modality is not necessarily the same as rejecting the interaction.* BrewBot therefore treats speech and gesture as substitutable channels rather than merely two interfaces available in parallel. The open palm functions as a form of modality negotiation: the human can determine not only the answer to the robot's question, but also how the conversation should proceed.

2.3 Embodied Drink Assistance

Once a beverage is explicitly accepted, interaction becomes physical. BrewBot combines stored task poses with motion planning to pick up a glass or cup, travel between stations using the ceiling rail, interact with available appliances, and deliver the resulting drink to a predefined location such as the user's desk.

Coffee and hot water for tea can be requested through the networked coffee machine via Home Assistant and Home Connect. The robot places the cup at the coffee machine, starts the drink program through Home Assistant, waits for dispensing, and then delivers the cup. The details of motion planning are deliberately secondary to the HRI design: from the user's perspective, the important property is that a conversational or gestural decision leads to a visible physical outcome in the shared environment.

Water requires a different interaction pattern because BrewBot cannot operate the tap itself. The robot picks up a glass, positions it under the faucet, and requests human assistance through the interaction system while holding the glass in place. The human opens the tap, fills the glass, and closes it again before signalling completion through speech or gesture. Rather than treating the inability to manipulate the faucet as a failure, the task becomes a small instance of *human–robot collaborative execution* in which each participant contributes the action they can perform more easily.

This interaction also exposed a problem of robot legibility. While waiting under the faucet, a stationary arm can be difficult to distinguish from a robot that has stopped functioning. BrewBot therefore uses staged non-verbal attention cues when the wait becomes prolonged: the arm first performs a small wrist movement while continuing to hold the glass, and then the smart lights blink to provide an additional environmental cue. These actions communicate system state without requiring repeated spoken interruptions.

The same principle motivates the robot's orientation during gesture interaction. Turning toward the user and adding visible questioning movements is not necessary merely for gesture recognition; it helps communicate that an interaction is active and that the next action belongs to the human. BrewBot thus uses physical motion both instrumentally, to manipulate objects, and communicatively, to make the interaction state more legible.

3 Design Reflections

BrewBot was developed as an integrated course prototype rather than as a controlled experiment. The following observations should therefore be interpreted as design reflections and hypotheses, not validated design principles.

3.1 Separate State Inference from Interruption Policy

An early architectural temptation in a state-aware robot is to let the inference component directly trigger behavior: if the system identifies a relevant state, it immediately sends a corresponding robot command. BrewBot instead separates state estimation from interaction initiation.

This separation matters because *what to offer* and *when to interrupt* depend on different information. Physiological measurements may suggest that water or another beverage could be appropriate, but they do not indicate whether the user is currently available for interaction. Arrival, recent offers, an ongoing dialogue, previous rejection, and the persistence of a sensor condition all influence whether initiating another interaction is sensible.

Keeping these responsibilities separate also makes different intervention policies possible without changing the state estimator. More sophisticated future policies could incorporate explicit availability cues, task context, learned interruption preferences, or confidence in the estimated state while retaining the same sensing pipeline.

3.2 Make Modalities Substitutable and Preserve User Agency

Simply supporting speech and gestures does not make an interaction meaningfully multimodal. BrewBot's more consequential design choice is that one channel can replace another while preserving the interaction state.

The open-palm mechanism illustrates this directly. The user can reject speech without rejecting the underlying beverage offer, after which the robot continues through physical gestures. Gesture meanings are furthermore interpreted relative to the current question, allowing the same limited vocabulary to function under different physical constraints.

This design is closely connected to human agency. BrewBot initiates interaction based on information the user did not deliberately provide as a command. Such implicit input should not silently become authorization for physical action. We therefore maintain the sequence:

> sensed cue → candidate state → suggestion → human confirmation → physical action

rather than:

> sensed cue → inferred state → physical action

The human can accept the suggestion, reject it, choose an alternative, change the communication modality, or terminate the interaction. In this sense, proactivity determines who may *start* the interaction, but not who has final authority over the resulting action.

4 Limitations

BrewBot was demonstrated and iteratively developed by the project team, but the system was not evaluated with independent participants in a user study. We therefore cannot conclude that its proactive behavior is perceived as useful rather than intrusive, that its robot gestures are intuitively understandable, or that allowing users to (seamlessly) switch modalities improves the interaction.

A second limitation concerns human-state inference. The current estimator is heuristic and has not been calibrated against self-reported ground truth. Physiological measures vary substantially between people and can be affected by movement, environment, sensor placement, and other factors unrelated to the states represented by BrewBot's labels. Although relative baselines and explicit handling of some physical confounds improve the rule set, the resulting categories remain tentative hypotheses rather than validated state estimates.

The mapping from states to beverages is similarly illustrative. A person with elevated arousal does not necessarily want tea, and a low-arousal signal does not imply that coffee is desirable. Beverage selection is therefore best understood as an adaptive interaction policy chosen for the prototype. A deployable system would need a preference model or recommender component that learn from repeated acceptance and rejection rather than apply one mapping to all users.

The physical system also relies strongly on a structured environment. Many poses and object locations are predefined, which limits robustness when cups or other objects are moved. The current interaction was tested over demonstration-length sessions rather than long-term deployment, so the acceptability of proactive offers over several hours or days remains untested.

Finally, physiological sensing raises privacy considerations beyond those of a conventional drink-serving robot. Signals such as heart rate and EDA constitute personal data and may permit inferences users did not explicitly intend to communicate. The prototype operates locally in a research environment, but long-term use would require explicit decisions about consent, transparency, storage, retention, and the user's ability to disable state-aware behavior.

5 Future Work

An empirical evaluation of the interaction design would be a useful next step, alongside further robotic capabilities. A within-subjects study could compare proactive state-aware assistance with user-initiated service, measuring perceived intrusiveness, usefulness, trust, usability, and task load. A second factor could compare speech-only interaction with the substitutable speech-and-gesture design. Such a study would directly test the hypothesis emerging from BrewBot that allowing users to replace an inconvenient interaction modality can make proactive assistance easier to manage.

Gesture comprehensibility is another open question. The human gestures used by BrewBot are intentionally simple, but the robot's beverage mimes are designed rather than universally established symbols. Future work should therefore test whether users can correctly interpret these gestures without prior instruction and whether different robot motions convey waiting, questioning, or offering states as intended.

Human-state inference should also become individualized and uncertainty-aware. A short calibration phase could establish personal baselines for physiological measures, while repeated interaction could allow the system to learn individual mappings between state, context, and beverage preference. Instead of treating a rule as "low arousal implies coffee," the system could learn that a particular user repeatedly chooses water or declines assistance in that context. Confidence estimates could additionally allow BrewBot to refrain from proactive interaction when the available sensor evidence is ambiguous.

Longer-term interaction also requires adaptive interruption policy. Repeated rejection should influence future behavior, and the system should learn both *what* a person tends to accept and *when* they prefer not to be interrupted. This would extend BrewBot from a state-aware system toward a genuinely personalized mixed-initiative assistant.

On the physical side, fiducial markers such as AprilTags [8] or more general object perception could reduce dependence on hard-coded poses and enable dynamic cup pickup and placement. Detecting whether a cup is present or empty could also provide task-context information that complements physiological sensing. Such environmental evidence may in many situations be more directly relevant to the need for a refill than a physiological cue alone.

6 Conclusion

BrewBot demonstrates an end-to-end approach to human-state-aware robotic assistance in which physiological sensing, proactive interaction, multimodal communication, smart-home integration, and embodied action are connected within one system. The central challenge that emerged was not simply how to infer a state or how to deliver a drink, but how a robot should negotiate assistance when it initiates the interaction itself.

Three design considerations summarize this experience. First, estimating *what* might be useful should be separated from deciding *when* interaction is appropriate. Second, multimodal interaction becomes more meaningful when modalities can substitute for one another according to the user's situation. Third, state inference should inform a proposal rather than remove human authority over the resulting action.

BrewBot therefore treats physiological state not as an instruction, but as a reason to ask. The robot may take the initiative, but the user decides whether the interaction continues, which modality it uses, and what the robot ultimately does.

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